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Hemocyte Apoptosis as a Biomarker of Environmental Stress in Aeglid Species (Decapoda: Anomura)

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Abstract
Neotropical aquatic ecosystems harbor a rich diversity of endemic species, many critically threatened by anthropic activities, such as the aeglid crabs. Hemolymph cell morphology and viability analyses offer diagnostic tools for evaluating the biological condition of species subjected to environmental stressors. This study investigates apoptotic cell ratios in freshwater crabs of the genus *Aegla*, exposed to anthropic impact. Hemocyte morphology and apoptosis levels were assessed in *Aegla* spp. collected from streams with different levels of disturbance across Brazil's southeast region. Water quality parameters were measured alongside concentrations of metals and surfactants. Sites exhibiting the lowest water quality presented significantly elevated apoptotic cell ratios. The integration of hemolymph analyses with environmental assessments revealed potential correlations between biotic and abiotic factors, offering an understanding of the relationship between immune status and ecological conditions in aeglid species. This integrative approach provides a valuable framework for informing conservation strategies for these freshwater organisms.

Keywords Apoptotic ratio · Flow cytometry · Immunology · Conservation · Decapoda

Introduction

The conservation status of threatened species is commonly assessed based on data regarding their spatial distribution, habitat quality, and occurrence patterns (MMA 2014; ICMBIO/MMA 2018; IUCN Standards and Petitions Committee 2022). Key biomarkers are increasingly used to evaluate the physiological responses of animals to environmental disturbances, making them essential tools for assessing population health in conservation biology (Beaulieu and Costantini 2014). Decapod crustaceans of the genus *Aegla* Leach, 1821 (Aeglidae Dana, 1852) are freshwater anomurans endemic to southern South America (Martin and Abele 1986; Pérez-Losada et al. 2004). Several of these species are subject to environmental pressures such as riparian forest degradation and the resulting sedimentation of rivers and lakes (Bond-Buckup et al. 2010). To date, 100 species have been described (Arantes et al. 2024; Bueno et al. 2024; Colavite et al. 2024; Arantes et al. 2025), and 18 of the 49 Brazilian species assessed (36.7%) are currently classified as threatened (ICMBio 2023), a situation that led to the species being included in the Ministry of the Environment's (MMA) "National Action Plan (PAN) for the Conservation

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of Endangered Atlantic Forest Fish and Aeglas".The primary threats to aeglids stem from activities that degrade the quality of their habitats, including the suppression of riparian forests, sedimentation of streams, and pollutants associated with agricultural, livestock, and aquaculture practices (Boos et al. 2020). These factors lead to habitat degradation and a subsequent decline in water quality, adversely affecting various aeglid species distributed in those environments (Pacca et al. 2007; Bueno et al. 2010, 2018; Santos et al. 2012; Boos et al. 2020). One practical approach to assessing the impact of pollutants on freshwater organisms and their immune function is the analysis of hemolymph, the fluid responsible for transporting nutrients, hormones, and metabolic waste products (Grubhoffer et al. 2013). The study of hemocytes, the immunocompetent cells in aeglids (Söderhäll et al. 1994; Guise et al. 2005), may provide valuable insights into the welfare of the animals (Xian et al. 2021).

Hemocytes in decapods are generally classified into three distinct morphological types: granulocytes, semi-granulocytes, and hyalinocytes, each associated with specific immune functions such as phagocytosis, cytotoxicity, and encapsulation (Bauchau 1981; Cárdenas et al. 2000; Matozzo and Marin 2010; Xian et al. 2012; Xing et al. 2017; Bandeira et al. 2020). One method for evaluating these cells is flow cytometry, a technique widely used to investigate cellular immunity and various biological processes, particularly cell death (Cárdenas et al. 2000; Galhardas 2014). Previous studies have demonstrated that exposure to varying concentrations of nitrite (Xian et al. 2011; Guo et al. 2013) and copper (Xian et al. 2010; Guo et al. 2017) can lead to increased levels of apoptotic cells in decapods. Given the critical conservation status of endemic decapods such as

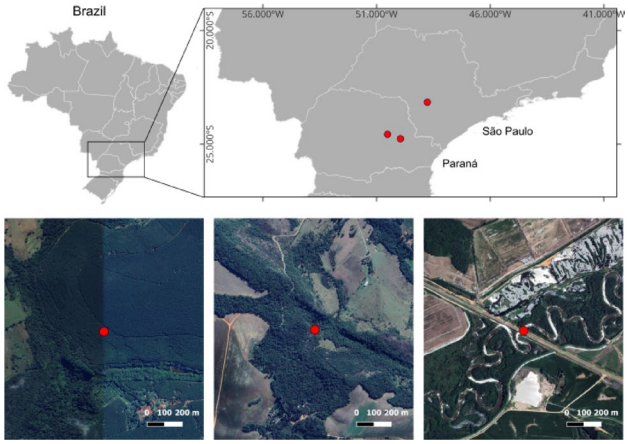

Fig. 1 Sampling sites in the states of São Paulo and Paraná, Brazil. On the left bottom: Pedra Preta Stream (PP), in the city of Avaré, site located in a transition between preserved vegetation and an area of monoculture; center bottom: Salto Puxa Nervos (PN), in the town of Tibagi, situated in an area of preserved forest, inside a particular farm, and right bottom: Iapó River (IR), in the city of Castro, site located below a highway and surrounded by areas of plantation

aeglids and their occurrence in aquatic environments subject to varying degrees of anthropogenic impact, this study aimed to analyze the apoptotic ratio of hemocytes in aeglid species by identifying live, early apoptotic, and dead cells; and correlate biotic and abiotic data from sampling sites to identify potential human impacts on those freshwater crustacean populations. A description of the morphology of the different hemocyte types found in *Aegla* sp. from Avaré—SP is also available. It was expected that the hemocytes of specimens collected from the more impacted sites would show a higher apoptosis/necrosis ratio compared to those from animals in the more preserved sites.

Materials and methods

Specimens and water samples were collected from three locations, corresponding to the previously documented geographic distribution of aeglids (Wolf 2018), during May 2023. One site in the Pedra Preta stream (PP, 23° 09′ S, 48° 46′ W; altitude 635 m), located in the municipality of Avaré, São Paulo State, Brazil; and two additional sites in the state of Paraná, Brazil: Salto Puxa Nervos (PN, 24° 56′ S, 50° 52′ W; altitude 785 m), municipality of Tibagi, and the Iapó River (IR, 24° 81′ S, 49° 85′ W; altitude 1001 m), municipality of Castro (Fig. 1). Specimens of *Aegla* spp. were collected through passive sampling using perforated plastic traps containing bait (cat feed), randomly deployed at the sampling sites during the late afternoon and retrieved the following morning (Bueno et al. 2007; Grabowski et al. 2013; da Silva et al. 2016), as aeglids exhibit heightened exploratory activity at night (Sokolowicz et al. 2007; De Almeida et al. 2021). Only intermolt animals from both sexes were used in this research, and they were transported to the laboratory (~90 min for Avaré and ~5 h for Paraná) in aerated plastic coolers with approximately 8 L of stream water (~8 animals per cooler, individualized in plastic pots). Forty-six specimens were obtained from three locations, being 18 from PP (Avaré), 19 from PN (Tibagi), and 9 *Aegla castro* Schmitt, 1942 from IR (Castro). No mortality was observed during and after the transportation.

Multiple sites along the Pardo River were identified as contaminated by Cu and Mg, including one in Avaré, likely resulting from anthropogenic activities linked to agricultural practices (Bonotto 2024). In the Paraná State, studies reported contamination by Cr, Cd, and Pb, pollutants associated with intensive farming and livestock activities in the rivers near the city of Castro (São João and Lajeado Carambeí rivers) (da Silva et al. 2017). In addition, da Silva et al. (2021) found high concentrations of metals in tributaries of the Tibagi River, attributed to the indiscriminate use of

agrochemicals, untreated sewage discharge, and fossil fuel combustion byproducts.

Hemolymph analyses: around 5 specimens per tank were maintained for 1 to 3-d, in aerated aquaria equipped with constant water circulation from the respective stream they were collected from (15 L), both natural (rocks) and artificial shelters, under temperatures of around 20 °C and 21 °C (monitored with a thermometer), a 12-h photoperiod (set with a timer), and filtration (Cogo et al. 2018). The tanks were renewed daily with water collected from the original sampling streams, as species from each location were acclimatized in different aquaria, and pH was monitored with pH stripes (ranging from 4.9 to 5.2). Hemolymph was extracted from the base of the fourth pereopod of each specimen (adapted from Gollas-Galván et al. 1997) using Alsever's anticoagulant solution (glucose 20.5 g L^{-1}, sodium citrate 8 g L^{-1}, sodium chloride 4.2 g L^{-1}, pH 7.5). Cell viability was quantitatively assessed by analyzing apoptotic hemocytes using flow cytometry, with the Annexin Apoptosis Detection Kit V-FITC/IP (BD Pharmingen, BD Biosciences, New Jersey, USA), following the procedures described by the manufacturer, and never surpassing 1×10^6 cells. Additionally, hemocytes from the processed samples were morphologically characterized through images from a microscope (Nikon Ni, Tokyo, Japan), projected through a CCD camera (DiSR2, Nikon, Tokyo, Japan), and captured with the Nis-Ar Elements software (Nikon, Tokyo, Japan).

Subsequently, a death control was established to calibrate the cytometer. All macrotubes containing the hemolymph-anticoagulant mixture were centrifuged and decanted. The resultant pellets were resuspended in the Alsever's solution/Annexin buffer (BD Pharmingen, BD Biosciences, New Jersey, USA) and stained with Annexin V-FITC and Propidium Iodide. After incubation, each tube was supplemented with Alsever's solution/Annexin buffer and analyzed by flow cytometry to determine viable, early apoptotic, and late apoptotic or dead cells (Van Engeland et al. 1998; Galhardas 2014; Braga et al. 2016).

Flow cytometric analyses were performed using a FACSCanto II flow cytometer (BD Biosciences; San Jose, CA, USA), with the FACSdiva software, equipped with filters for fluorescence detection and histogram generation. Data acquisition and the construction of dot plots were conducted using FlowJo software, version vX.10.6 (Tree Stars, Ashland, OR, USA). The apoptotic ratios obtained from the three sampling sites were compared with corresponding water quality parameters to evaluate potential correlations between environmental conditions and the proportions of viable and non-viable hemocytes. The Shapiro–Wilk test indicated a normal distribution for the apoptosis rate data ($P = 0.35$). The Levene's test confirmed the homoscedasticity of the data ($P = 0.40$). Subsequently, a one-way analysis of variance (ANOVA) was performed, revealing significant differences in apoptosis rates among the three sampling sites ($P < 0.05$). A Tukey multiple range post hoc test was then conducted to identify which groups differed significantly.

Water samples were collected and refrigerated from each site using 1-L decontaminated containers. Metal concentrations (Cu, Zn, Pb, Cr, Ni, Cd, and Hg) were determined using graphite furnace atomic absorption spectrometry (GFAAS) with a SHIMADZU AA-6800 spectrometer. A pyrolytic graphite tube with an integrated tungsten-coated platform and a deuterium background correction system was used. Palladium (Pd^{2+}) was applied as a chemical modifier for most metals, while zirconium (Zr^{4+}) was used specifically for Hg. Specific furnace temperature programs were employed for each metal, and absorbance was measured using peak area readings (Leite et al. 2019; Bittarello et al. 2020). The calculated limit of detection (LOD) was 0.001 µg/g. Validation was performed by analyzing percentage metal recovery from the certified reference material (> 80%). Subsequently, they were analyzed using the parameters defined by the IPMCA (Index of Minimum Variables for the Preservation of Aquatic Life) (CETESB 2021).

Environmental quality levels are represented by numerical scores ranging from 1 to 3, corresponding to water quality standards from best to worst. These classifications are based on the guidelines established by CONAMA Resolution No. 357/2005 and environmental standards recommended by the U.S. (USEPA 1991) and French (Code Permanent: Environment and Nuisances 1986) legislation. Acute and chronic ecotoxicity tests were conducted using the NBR 13373:2017 standard (ABNT 2017), using the microcrustacean *Ceriodaphnia silvestrii* as a test organism by evaluating neonate mortality (acute toxicity) and reproduction (chronic toxicity) in the various water samples. The analysis were considered valid, as they presented less than 20% of mortality in the control samples. For surfactant analysis, the 5540 C method from the Standard Methods for the Examination of Water and Wastewater (SMEWW 2017) was employed to quantify anionic surfactants, expressed as Methylene Blue Active Substances (MBAS).

The Carlson's Trophic State Index (IET), as modified by Lamparelli (2004), was also evaluated. The final IET value was interpreted using a classification table (Table 1) and converted into numerical scores incorporated into the Water Quality Index for the Protection of Aquatic Life and Aquatic Communities (IVA). Water pH, dissolved oxygen, and chlorophyll concentrations were monitored using a multiparameter probe (Eureka MANTA 2) during field campaigns conducted in May 2023. The results obtained from these analyses were compared with reference tables provided in the Inland Water Quality Report of the State of São Paulo (CETESB 2021) to assess contamination levels. The

Table 1 Classification of the trophic state for rivers according to the modified Carlson Index (adapted from CETESB 2021).

Category (Trophic state)	Ponderation	Total-P – P (mg m^{-3})	Chlorophyll-a (mg m^{-3})
Ultraoligotrophic	IET≤47	P≤13	CL≤0.74
Oligotrophic	47<GO≤52	13<P≤35	0.74<CL≤1.31
Mesotrophic	52<GO≤59	35<P≤137	1.31<CL≤2.96
Eutrophic	59<GO≤63	137<P≤296	2.96<CL≤4.7
Supereutrophic	63<GO≤67	296<P≤640	4.7<CL<7.46
Hypereutrophic	IET>67	640<P	7.46<CL

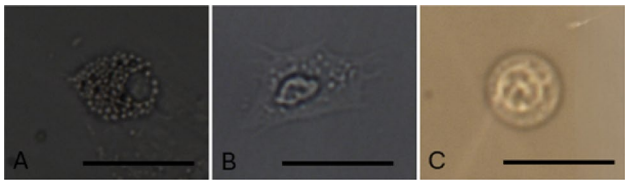

Fig. 2 Hemocytes found in the hemolymph of *Aegla* specimens collected in Avaré—SP. Images taken in a Neubauer chamber with the cells in vivo. **A** Granulocyte. **B** Semigranulocyte. **C** Hyalinocytes. Scale bar: 50 µm

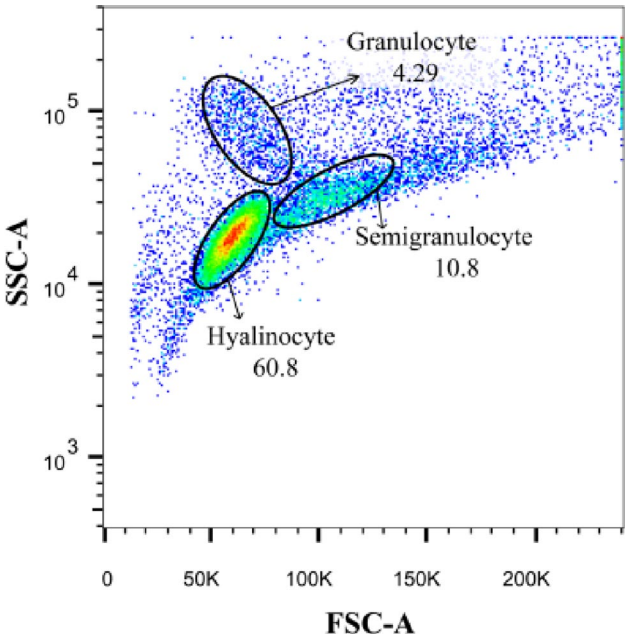

Fig. 3 Dot plot of the 3 types of *Aegla* sp. hemocytes collected from the Pedra Preta Stream in Avaré—SP. The FSC-A (forward scatter height) axis represents cell size, while the SSC-A (size scatter height) axis represents granularity. Red, yellow and green represent a large concentration of cells, as opposed to blue

Water Quality Index for the Protection of Aquatic Life and Aquatic Communities (IVA) was subsequently calculated, and its values were evaluated based on the corresponding IVA classification table. A Principal Component Analysis (PCA) was performed to classify the sampling locations based on abiotic factors. A matrix comprising all measured variables was constructed, normalized, and centered before analysis. All statistical procedures were conducted using R software, version 4.4.2 (R Core Team 2024).

Results and Discussion

Three distinct types of hemocytes were identified in the hemolymph of the specimens: granulocytes, characterized by an abundance of small cytoplasmic granules, typically drop-shaped, with a small, non-centralized circular nucleus (Fig. 2A); semigranulocytes, which are larger cells containing fewer granules, cytoplasmic projections, and a relatively large, ovoid, non-centralized nucleus (Fig. 2B); and hyalinocytes, smaller, circular cells lacking cytoplasmic granules and containing a large, central spherical nucleus (Fig. 2C). These three hemocyte types were also distinguished using flow cytometry, based on cell size (FSC-A) and granularity (SSC-A) parameters (Fig. 3).

Forty-six individuals were analyzed, with the hemolymph from one specimen per locality used as a death control in flow cytometry. Rates of early apoptosis, late apoptosis, and necrosis were represented in the dot plots by hemocytes located in quadrants B, C, and D, respectively (Fig. 4). Specimens collected from IR (Castro) exhibited an average apoptosis plus necrosis rate of 29.12% (sum of quadrants B, C, and D). *Aegla* sp. from PN (Tibagi) showed a rate of 34.09%, while *Aegla* sp. from PP (Avaré) presented the highest rate at 42.47% (Fig. 4). The specimens from PP (Avaré) exhibited the highest combined apoptosis/necrosis ratio compared to those from PN (Tibagi) and IR (Castro). A significant difference was observed between PP and IR (P=0.002) (Fig. 5).

The environmental parameters defined by the IPMCA for all three sampling sites were compiled into a table (Table 2). Concentration values were compared against CETESB reference thresholds to assign corresponding scores (ponderations). For dissolved oxygen, toxicity, and toxic substances, except for Cr, which received a score of 2, all sampling sites were assigned a ponderation of 1. Regarding pH, the PN (Tibagi) and IR (Castro) sites received a ponderation of 2, while PP (Avaré) was assigned a ponderation of 3. Environmental parameters related to the Trophic State Index (IET) were also compiled (Table 3), including data from all three sampling environments, and adjusted to provide ponderations for use in the calculation of the Water Quality Index for the Protection of Aquatic Life (IVA). The IVA scores indicated that PP (Avaré) had poor water quality (5.6), while PN (Tibagi) and IR (Castro) were classified as regular (both

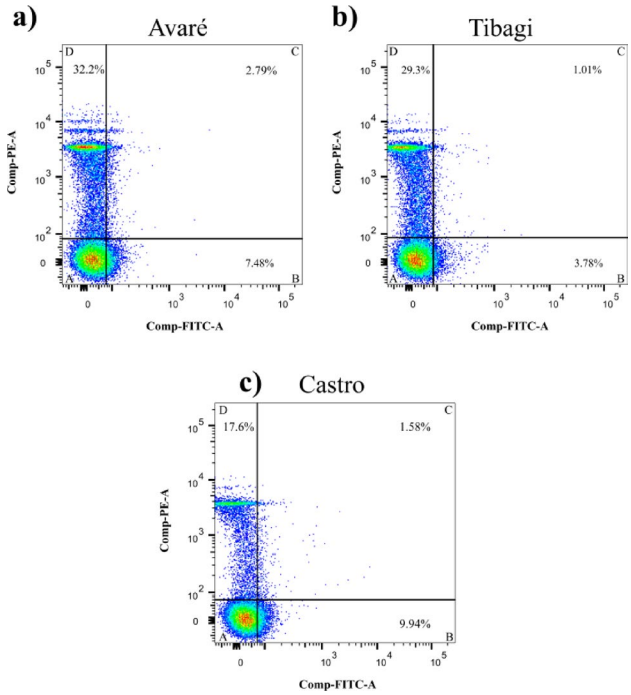

Fig. 4 Cytometry analysis of the hemocytes of the collected Aegla spp. specimens, stained with Annexin V-FITC (Comp-FITC-A) and Propidium Iodide (Comp-PE-A), and the respective percentages of cells in each quadrant. **a***Aegla* sp. from Pedra Preta, Avaré. **b***Aegla* sp. from Puxa Nervo, Tibagi. **c***Aegla castro* de Iapó river, Castro. Live cells (FITC-A-/PE-A-) are in quadrant A; cells in early stages of apoptosis (FITC-A+/PE-A-), in quadrant B; cells in the final stages of apoptosis (FITC-A+/PE-A+), in quadrant C, and cells in necrosis (FITC-A-/PE-A+), in quadrant D. Red, yellow and green represent a large concentration of cells, as opposed to blue.

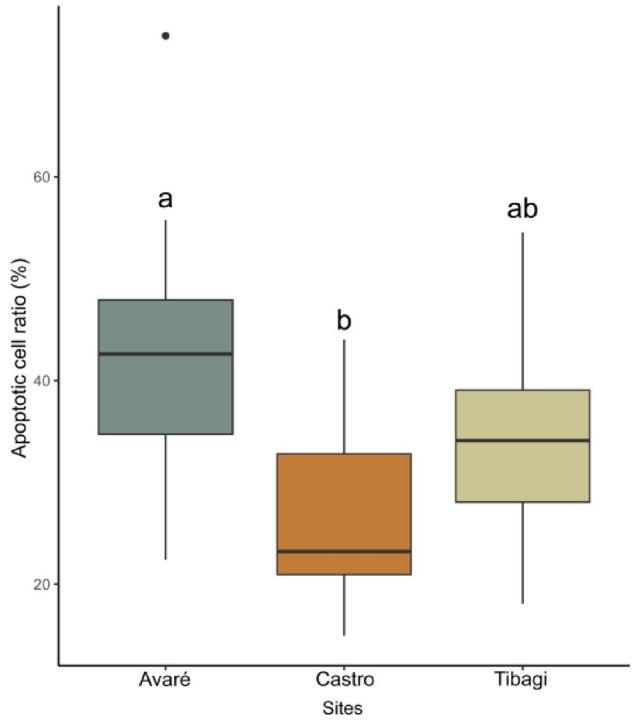

Fig. 5 Boxplot represents the medians of the apoptosis rates of Avaré (Pedra Preta—PP), Castro (Iapo river—IR), and Tibagi (Puxa Nervo—PN) specimens. The distribution indicates a difference between the apoptosis rates of Avaré and Castro and similarities between the apoptosis rates of Avaré and Tibagi and between Castro and Tibagi. Different letters are significantly different at Tukey tests, $p < 0.05$

4.4). These scores were obtained using reference values, as well as classification criteria of quality of water pre-established by the São Paulo state Environmental Company (CETESB 2021) and available as supplementary material of this article (Online Resource 1 and Online Resource 2). Principal component analysis (PCA) revealed that the first two components explained approximately 92% of the total variance, with PC1 accounting for 55.09%. The apoptotic ratio positively correlated with PC1, while pH was negatively correlated, suggesting an association between these

Table 2 Description of the water parameters of the IPMCA (Index of minimum variables for the preservation of aquatic life) analyzed to determine the IVA (Water quality index for the protection of aquatic life and aquatic Communities) of each environment

Abiotic factors	Pedra Preta—Avaré	Puxa Nervos—Tibagi	Iapó river—Castro	International parameters
DO (mg/L)	9.2	8.3	5.5	US EPA: > 5.5 CCME: > 5.5
pH (Sorensen)	4.9	5.0	5.1	US EPA: 6.5–9.0 CCME: 6.5–9.0
Toxicity	Non-toxic	Non-toxic	Non-toxic	–
Cd (mg/L)	n.a.	n.a.	n.a.	US EPA: 1.8×10^{-3} CCME: 9.0×10^{-5}
Cr (mg/L)	0.1683	0.0922	0.1483	US EPA: 0.074 CCME: 0.0089
Dissolved Cu (mg/L)	0.0064	0.0048	0.0037	–
Total Pb (mg/L)	n.a.	n.a.	n.a.	US EPA: 0.0025
Hg (mg/L)	n.a.	n.a.	n.a.	US EPA: 7.7×10^{-4} CCME: 1.0×10^{-6}
Ni (mg/L)	0.0038	0.0037	0.0038	US EPA: 0.052
Surfactants (mg/L)	< 0.01	< 0.01	< 0.01	–
Zn (mg/L)	0.1028	0.0644	0.1097	US EPA: 0.12 CCME: 0.007

Table 3 Description of the water parameters of the IET (Trophic state index) analyzed to determine the IVA (Water quality index for the protection of aquatic life and aquatic Communities) of each environment

Site	Chlorophyll-a µg/L	Total *P* µg/L	IET chlorophyll-a	IET *P*	IET total	Category (trophic state)
Pedra Preta—Avaré	0.21	179.53	45.97	60.90	53	Mesotrophic
Puxa Nervo—Tibagi	0.72	414.1	47.26	65.24	56	Mesotrophic
Iapó river—Castro	1.41	351.3	53.07	64.38	59	Mesotrophic

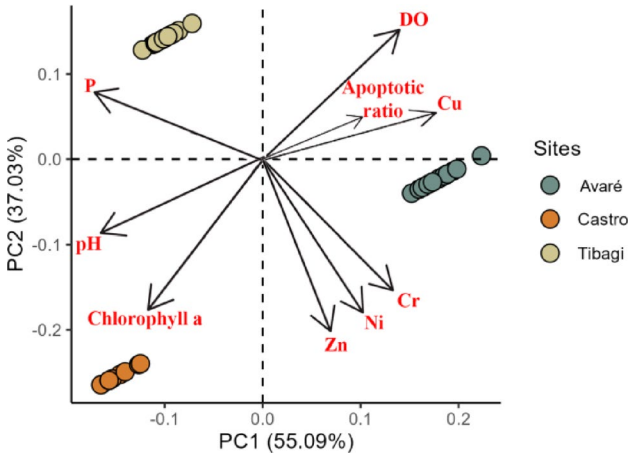

Fig. 6 Principal Components Analysis of pH, Dissolved Oxygen (DO), Phosphorus (P), Chlorophyll-a, and correlations with Aegla's Apoptotic ratio on all 3 sites sampled

variables. Copper also correlated positively with PC1, whereas phosphorus showed a negative correlation. PC2 explained 37.03% of the variance and was negatively correlated with chlorophyll-a and the other metals: Zn, Cr, and Ni (Fig. 6).

The primary threats faced by aeglids include the suppression of riparian forests, siltation, and the contamination of water bodies resulting from agricultural, livestock, and aquaculture activities, particularly in the Atlantic Forest, where most aeglid species in Brazil are distributed (Boos et al. 2020). This biome retains only about 12% of its original extent (Ribeiro et al. 2009). It is recognized as a biodiversity hotspot (Rezende et al. 2018), which prompted the establishment of the Ministry of the Environment's "National Action Plan for the Conservation of Atlantic Forest Fish and Aeglids". Consequently, pollution by metals originating from agricultural practices, such as the use of fertilizers, pesticides, livestock manure, and wastewater (Alengebawy et al. 2021), as well as the application of fungicides and feed additives (Tamm et al. 2022), and phosphorus-based fertilizers responsible for eutrophication (Bechmann and Stålnacke 2019), could be important parameters to consider when evaluating the impacts of environmental pollution on fauna inhabiting these regions, as well as the ecological risk of these environments.

Consistent with the Water Quality Index for the Protection of Aquatic Life and Aquatic Communities (IVA), higher apoptotic ratios were observed in hemocytes from individuals collected at the site with the least favorable conditions for aquatic life. This finding suggests that various factors, such as acidic pH, dissolved oxygen concentrations, and potential immunological differences among aeglid species, may influence specific biomarkers in these specimens (Correa-Araneda et al. 2010; Dalosto and Santos 2011; Kawamura et al. 2015; Tomasetti et al. 2018). Investigating O_2 consumption in Neotropical freshwater decapods, Dalosto and Santos (2011) found that *Trichodactylus panoplus* von Martens, 1869 and *Parastacus brasiliensis* (von Martens, 1869) modulated their metabolism and O_2 consumption according to its availability on the environment, while *Aegla longirostri* depended on well-oxygenated waters because of its constantly high metabolic rates. Borges et al. (2022) reported that Cd, Cr, and Fe sediment concentrations affected oxidative biomarkers in aeglids, prompting adaptive responses to cope with stress. Exposure to Cd led to a decrease in the total hemocyte counts of the freshwater crab *Sinopotamon henanense*, as well as oxidative stress, and damage to the mitochondria, nucleus and endoplasmic reticulum of the hemocytes (Qin et al. 2012; Zhou et al. 2016).

The most prominent environmental factor was the low pH in PP (Avaré), which coincided with the highest levels of hemocyte apoptosis, suggesting a potential link between acidity and cellular stress. Acidic pH has been identified as a limiting factor for crustaceans, negatively impacting the distribution of amphipods (Stephenson and Mackie 1986; Taylor et al. 1994) and crayfish (Berrill et al. 1985), as well as affecting survival, growth, and exoskeleton development in caridean shrimp *Macrobrachium rosenbergii* (Kawamura et al. 2015), and growth and reproduction in ostracod *Heterocypris incongruens* (Ramdohr, 1808). Slightly acidic pH levels (between 6.2 and 6.6) have also proven unfavorable for the presence of *Aegla rostrata* Jara, 1977, a Chilean species (Correa-Araneda et al. 2010). In freshwater environments, where calcium carbonate ($CaCO_3$), one of the main components of crustacean exoskeletons, is not as abundant as in the ocean, these organisms rely on a process called calcification, which requires alkaline conditions, to form $CaCO_3$, and therefore, their protective structures (Wood and Cameron 1985; Whiteley 2011; Kim et al. 2015). In a review on the effects of freshwater acidification, Muniz (1990) reported a consistent negative correlation between pH levels and the species richness and diversity of benthic

macroinvertebrates, even under moderately acidic conditions ranging from pH 5.0 to 6.0. Some pollutants can interfere with the total hemocyte count (THC), leading to its reduction as hemocytes begin to undergo apoptosis. For instance, sulfide can reduce phenoloxidase activity, impairing the recognition of exogenous particles and hindering infection prevention, while increasing superoxide anion levels, thereby inducing cytotoxicity (Cheng et al. 2007). Similarly, Cu can elevate the production of reactive oxygen species (ROS), contributing to oxidative damage (Xian et al. 2010; Guo et al. 2017), and nitrite can also trigger oxidative stress and cytotoxic effects (Xian et al. 2011, 2012; Guo et al. 2013), as well as metals like Cd and Hg (Frías-Espericueta et al. 2022). However, the analysis of the presence and concentration of toxic substances in the water indicated conditions favorable for the survival of aquatic organisms across all three environments, except for Cr, which was found at concentrations potentially harmful to reproduction in the long term (CETESB 2021). Furthermore, no evidence of toxicity to aquatic organisms was reported in the water samples (ABNT 2017), suggesting that, regarding these parameters, the environments generally exhibited suitable conditions.

The total P levels in the water were particularly noteworthy, as all sampling sites exhibited eutrophic or hypereutrophic concentrations, according to the Trophic State Index (IET). The application of N and P based fertilizers in agricultural and livestock activities, followed by their runoff into river systems, may contribute to these elevated values (Howarth et al. 2002; Mainstone and Parr 2002), especially considering that all three sites were in areas with surrounding cattle farming. Tibagi (PN) and Castro (IR) cities are situated in the northeastern region of the state of Paraná, which is characterized by extensive pasturelands, mining activities, livestock farming, human settlements, industrial hubs, and reforestation zones (Fasolo et al. 2002; Wolf 2018). Similarly, the stream in PP (Avaré) is surrounded by agricultural properties, further suggesting a potential link between land use and nutrient enrichment in these aquatic systems. Despite the elevated P levels, the eutrophication process is more difficult to establish in stream environments, as the continuous water flow facilitates the diffusion of atmospheric oxygen into the water column (Dodds 2006). Even with high P concentrations, all sites analyzed exhibited optimal dissolved oxygen levels. Nonetheless, IR was classified as a mesotrophic, nearly eutrophic environment, which, combined with lower water circulation compared to the other sites, may explain the relatively reduced dissolved oxygen levels observed. Limited water movement diminishes the atmospheric exchange of O_2 (Dodds 2006), while eutrophication further exacerbates the issue by inhibiting photosynthesis below the surface and increasing aerobic decomposition, leading to greater O_2 consumption (Khan and Ansari 2005).

In addition to the environmental differences observed, distinct taxa were also analyzed. Although aeglids exhibit relatively similar morphological traits across species (Jara et al. 2003), each species may present physiological particularities that could help explain the varying apoptosis ratios observed. For example, the three main hemocyte types commonly described in decapods: hyaline, semigranular, and granular, were identified, as previously reported for *Aegla castro* (da Silva Rosa et al. 2024). However, whereas hyalinocytes were the most abundant in the hemolymph of *A. castro* in that study, semigranulocytes were more prevalent in the present analysis. This variation may correlate with pollutant exposure, as Cu-induced stress has increased both total hemocyte count and the proportion of hyaline hemocytes in *A. castro* (da Silva Rosa et al. 2024). While controlled laboratory studies are necessary to isolate specific causal relationships, our findings indicate that hemocyte apoptosis ratios show promise as a supplementary biomarker for assessing stress in crustaceans in situ, enhancing current methods for evaluating the conservation status of these species. These data are innovative and have potential applications in both basic and applied sciences.

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